

Mean-centroid drift detection: null distribution and analytical calibration

Legal Word Drift — internal note

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1 Setup

Let P be a distribution on $\mathbb{R}^d$ with mean μ and covariance Σ . Under the null hypothesis H_0 (“no drift between the two groups”), all training and test samples are iid from P :

- Training: $X_1, \dots, X_n$ in the X -class, $Y_1, \dots, Y_m$ in the Y -class, all iid $\sim P$ and mutually independent.
- Test: $X'_1, \dots, X'_{n'}$ and $Y'_1, \dots, Y'_{m'}$, iid $\sim P$, independent of training.

Define

$$\hat{\Delta} = \bar{X}_{\text{tr}} - \bar{Y}_{\text{tr}}, \quad c = \frac{1}{2}(\bar{X}_{\text{tr}} + \bar{Y}_{\text{tr}}),$$

and classify a test point $z \in \mathbb{R}^d$ as X -class iff the score

$$S(z) := (z - c) \cdot \hat{\Delta} > 0.$$

2 Moments of $\hat{\Delta}$ under H_0

By linearity and independence of $\bar{X}_{\text{tr}}$ and $\bar{Y}_{\text{tr}}$,

$$\mathbb{E}[\hat{\Delta}] = \mu - \mu = 0, \quad \text{Cov}(\hat{\Delta}) = \frac{\Sigma}{n} + \frac{\Sigma}{m} = \left(\frac{1}{n} + \frac{1}{m}\right)\Sigma.$$

For the squared norm,

$$\mathbb{E}\|\hat{\Delta}\|^2 = \text{tr}(\text{Cov}(\hat{\Delta})) + \|\mathbb{E}\hat{\Delta}\|^2 = \left(\frac{1}{n} + \frac{1}{m}\right)\text{tr}(\Sigma).$$

For isotropic $\Sigma = \sigma^2 I_d$ and balanced $n = m = N$ this gives $\mathbb{E}\|\hat{\Delta}\|^2 = 2\sigma^2 d/N$, so the chord between the two centroids has length $\|\hat{\Delta}\| \sim \sigma\sqrt{d/N}$ even under H_0 . The centroid distance is not a calibrated test statistic without knowledge of (or an estimator for) Σ ; see [1] for the limiting geometry as $d \rightarrow \infty$ with n fixed.

3 Expected accuracy under H_0

Assume $\Pr_{X' \sim P}(S(X') = 0) = 0$ (e.g. P has a density). Under H_0 and *no other assumptions* on P , n, m, n', m' ,

$$\mathbb{E}[\text{accuracy} \mid H_0] = \frac{1}{2}.$$

Proof. Consider the relabeling that swaps every X -instance with a Y -instance, applied to the training data only. Call this map π . Under π ,

$$\bar{X}_{\text{tr}} \longleftrightarrow \bar{Y}_{\text{tr}}, \quad \hat{\Delta} \mapsto -\hat{\Delta}, \quad c \mapsto c,$$

so for any test point z the score flips sign: $S(z) \mapsto -S(z)$. Under H_0 the joint distribution of $(X_{\text{tr}}, Y_{\text{tr}})$ is invariant under π , because both training groups are iid from the same P . Hence the law of $S(z)$ conditional on H_0 equals the law of $-S(z)$: $S(z)$ is symmetric around 0 in distribution. With the no-atom-at-zero assumption,

$$\Pr(S(z) > 0) = \Pr(S(z) < 0) = \frac{1}{2}.$$

Since this holds for every test point regardless of its true class, $\mathbb{E}[\text{accuracy}] = 1/2$. $\square$

This uses no Gaussianity, no symmetry of P , no balance $n = m$. It is the same exchangeability machinery underlying permutation tests [3].

4 Variance of accuracy under H_0

Write $T = (X_{\text{tr}}, Y_{\text{tr}})$ and define the conditional accuracy on a single side,

$$p := \Pr(S(X') > 0 \mid T), \quad X' \sim P, \quad X' \perp T.$$

By Proposition 3, $\mathbb{E}[p] = 1/2$. Conditional on T , the $N_{\text{te}} = n' + m'$ test indicators are independent Bernoulli(p) (or $1 - p$, symmetric on the other side), and

$$\text{Var}(\text{accuracy}) = \mathbb{E}\left[\frac{p(1-p)}{N_{\text{te}}}\right] + \text{Var}(p) = \frac{1}{4N_{\text{te}}} + \text{Var}(p)\left(1 - \frac{1}{N_{\text{te}}}\right).$$

The first term is the irreducible finite-test binomial noise; the second is the spread of conditional accuracy across realisations of the training set. So

$$\text{Var}(\text{accuracy} \mid H_0) \geq \frac{1}{4N_{\text{te}}},$$

with equality only if $p \equiv 1/2$ almost surely. In high dimension ($d \gg n, m$) the inflation $\text{Var}(p)$ is non-negligible; see [2] for the high-dimensional regime in which fitted linear discriminants degenerate and centroid rules dominate.

5 AUROC: a fully analytical null

Replace the midpoint threshold by the rank statistic. Let

$$U = \sum_{i=1}^{n'} \sum_{j=1}^{m'} \mathbf{1}\{S(X'_i) > S(Y'_j)\}, \quad \widehat{\text{AUROC}} = \frac{U}{n' m'}.$$

Under H_0 (with P continuous), the conditional law of U given T is the standard Mann–Whitney null. In particular

$$\mathbb{E}[\widehat{\text{AUROC}} \mid T, H_0] = \frac{1}{2}, \quad \text{Var}(\widehat{\text{AUROC}} \mid T, H_0) = \frac{n' + m' + 1}{12 n' m'},$$

both *independent of T* .

Proof. Condition on T (hence on $\hat{\Delta}$ and c). The scores $\{S(X'_i)\}_{i=1}^{n'}$ and $\{S(Y'_j)\}_{j=1}^{m'}$ are then iid samples from the *same* univariate distribution: the pushforward of P through $z \mapsto S(z)$. (Under H_0 both test groups have source distribution P , and S is a fixed function once T is fixed.) The combined sample of $n' + m'$ scores is therefore exchangeable, and the rank of any subset of size n' has the Mann–Whitney null distribution [4], whose mean and variance are as stated and depend only on (n', m') . $\square$

For balanced $n' = m' = N$, $\text{Var}(\widehat{\text{AUROC}} \mid H_0) = (2N + 1)/(12N^2) \approx 1/(6N)$. There is no “between-classifier” inflation, because under H_0 the two test groups have identical projected distributions regardless of which direction $\hat{\Delta}$ the training data happens to pick out.

5.1 Derivation of the Mann–Whitney null variance

We derive $\text{Var}(U \mid T, H_0) = n' m' (n' + m' + 1)/12$ from first principles. Condition throughout on T ; under H_0 the test scores $\{S(X'_i)\}_{i \leq n'}$ and $\{S(Y'_j)\}_{j \leq m'}$ are then iid from a common continuous distribution F . Define

$$W_{ij} = \mathbf{1}\{S(X'_i) > S(Y'_j)\}, \quad U = \sum_{i=1}^{n'} \sum_{j=1}^{m'} W_{ij}.$$

Continuity of F gives $\Pr(S(X'_i) = S(Y'_j)) = 0$, so $\mathbb{E}[W_{ij}] = 1/2$ and $\text{Var}(W_{ij}) = 1/4$.

Expand

$$\text{Var}(U) = \sum_{i,j} \text{Var}(W_{ij}) + \sum_{(i,j) \neq (k,\ell)} \text{Cov}(W_{ij}, W_{k\ell}).$$

The diagonal contributes $n' m' \cdot \frac{1}{4}$. The off-diagonal covariances fall into three cases.

Case A ($i = k, j \neq \ell$). With $A = S(X'_i)$, $B = S(Y'_j)$, $C = S(Y'_\ell)$ iid $\sim F$,

$$\mathbb{E}[W_{ij}W_{i\ell}] = \Pr(A > B, A > C) = \Pr(A = \max\{A, B, C\}) = \frac{1}{3},$$

by exchangeability of three iid continuous random variables. Hence $\text{Cov}(W_{ij}, W_{i\ell}) = 1/3 - 1/4 = 1/12$. There are $n' \cdot m'(m' - 1)$ such ordered pairs.

Case B ($i \neq k, j = \ell$). By the symmetric argument with $A = S(X'_i)$, $B = S(X'_k)$, $C = S(Y'_j)$:

$$\mathbb{E}[W_{ij}W_{kj}] = \Pr(A > C, B > C) = \Pr(C = \min\{A, B, C\}) = \frac{1}{3},$$

so again $\text{Cov} = 1/12$, with $m' \cdot n'(n' - 1)$ ordered pairs.

Case C ($i \neq k, j \neq \ell$). Then W_{ij} and $W_{k\ell}$ depend on four disjoint iid scores, so they are independent and the covariance is 0.

Sum.

$$\begin{aligned} \text{Var}(U) &= n' m' \cdot \frac{1}{4} + n' m' (m' - 1) \cdot \frac{1}{12} + n' m' (n' - 1) \cdot \frac{1}{12} \\ &= \frac{n' m'}{12} [3 + (m' - 1) + (n' - 1)] = \frac{n' m' (n' + m' + 1)}{12}. \end{aligned}$$

Dividing by $(n' m')^2$ gives

$$\text{Var}(\widehat{\text{AUROC}} \mid T, H_0) = \frac{n' + m' + 1}{12 n' m'}.$$

None of the steps used the form of F beyond continuity, so the result is independent of T (as claimed) and the law of total variance collapses: $\text{Var}(\widehat{\text{AUROC}} \mid H_0)$ equals the same quantity unconditionally.

Normal approximation. For moderate n', m' a CLT for two-sample U -statistics gives $\widehat{\text{AUROC}} \mid H_0 \approx \mathcal{N}(\frac{1}{2}, (n' + m' + 1)/(12n'm'))$, so a one-sided z -test rejects H_0 at level α when

$$\widehat{\text{AUROC}} > \frac{1}{2} + z_{1-\alpha} \sqrt{\frac{n' + m' + 1}{12 n' m'}}.$$

For $n' = m' = 50$ this critical value at $\alpha = 0.05$ is $0.5 + 1.645 \cdot \sqrt{101/30000} \approx 0.595$. For small samples use the exact Mann–Whitney distribution [4].

6 Why fitted logistic regression breaks the argument

The label-swap proof of Proposition 3 relies on the direction $\hat{\Delta}$ being a function only of the two class means. For a fitted classifier (logistic regression, linear SVM) the trained direction depends on individual training points and is optimised to separate them; under H_0 it locks onto spurious finite-sample directions whose generalisation error is not $1/2$ in high dimension [2]. Consequently both $\mathbb{E}[\text{accuracy}]$ and $\mathbb{E}[\widehat{\text{AUROC}}]$ exceed $1/2$ under H_0 , and an analytical null no longer applies — one needs permutation calibration.

7 Practical recipe

For each year-pair $(t, t + 1)$ in the drift pipeline:

1. Split usages of each year into train / test (balanced if possible).
2. Compute $\hat{\Delta} = \bar{X}_{t+1, \text{tr}} - \bar{X}_{t, \text{tr}}$.
3. Project all test embeddings onto $\hat{\Delta}$ and compute $\widehat{\text{AUROC}}$.
4. Compare to the Mann–Whitney null: $z = (\widehat{\text{AUROC}} - 1/2) \cdot \sqrt{12 n' m' / (n' + m' + 1)}$.
5. Optionally report $\|\hat{\Delta}\|$ as a magnitude side-quantity, but calibrate it via permutation (or against $\text{tr}(\Sigma)$ estimated from the pooled sample), not analytically.

References

- [1] P. Hall, J. S. Marron, A. Neeman. Geometric representation of high-dimension, low-sample-size data. *Journal of the Royal Statistical Society, Series B*, 67(3):427–444, 2005.
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